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Chiral Complexes with *n*-Butyllithium and Methylzinc: X-ray Crystal Structures of Lithium and Zinc (1*R*,2*R*,4*S*)-2-*endo*-Oxido-2-*exo*-(*o*-methoxyphenyl)-1,3,3- trimethylbicyclo[2.2.1]heptane

Bernd Goldfuss,^{*,†} Saeed I. Khan,^{‡,§} and K. N. Houk[‡]

Organisch-Chemisches Institut der Universität Heidelberg, Im Neuenheimer Feld 270,
D-69120 Heidelberg, Germany, and Department of Chemistry and Biochemistry, University of
California, Los Angeles, 405 Hilgard Avenue, Los Angeles, California 90095-1569

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Summary: The ligand (1*R*,2*R*,4*S*)-2-*endo*-hydroxy-2-*exo*-(*o*-methoxybenzene)-1,3,3-trimethylbicyclo[2.2.1]-heptane (**1**) serves as a "chiral *n*-butyllithium trap" and precipitates *n*-butyllithium in complex **2**. With dimethylzinc, ligand **1** forms a dimeric zinc chelate complex (**3**). The X-ray crystal structures of **2** and **3** are discussed.

Complexes of organometallics with chiral ligands play a major role in asymmetric synthesis.¹ Organolithiums² add to prochiral carbonyl functions enantioselectively³ in the presence of chiral alkoxides⁴ or amines.⁵ Organozinc reagents enable the use of even catalytic quanti-

ties of chiral ligands in highly enantioselective alkylations.⁶ The molecular structures of these chiral intermediates can provide explanations for their reactivity and selectivity,⁷ but relatively few structures of chirally modified alkyllithium and alkylzinc reagents are known. Williard et al. reported X-ray structures of mixed

[†] Organisch-Chemisches Institut der Universität Heidelberg. E-mail: Bernd.Goldfuss@urz.uni-heidelberg.de.

[‡] University of California.

[§] Send correspondence concerning X-ray crystal analyses to this author.

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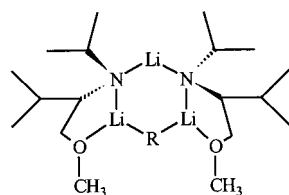
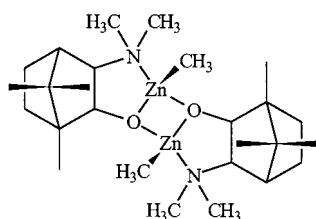
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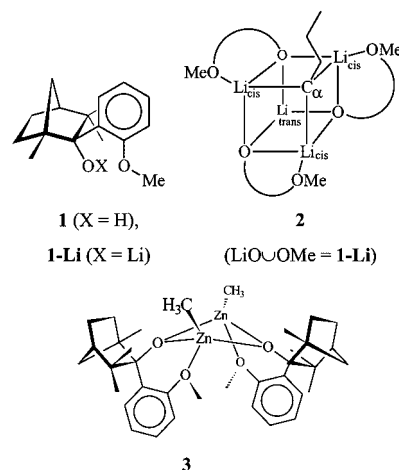
Scheme 1. Williard's Mixed Chiral Organolithium Aggregates (R = *n*-Bu, *s*-Bu, *t*-Bu)**Scheme 2. Noyori's Homochiral Syn Methylzinc Dimer of 3-*exo*-(Dimethylamino)isonorborneol (DAIB)**

aggregates of *n*-, *sec*-, and *tert*-butyllithium and *N*-isopropyl-*O*-methylvalinol (Scheme 1).⁸

Extensive studies by Noyori et al. have shown that monomeric zinc chelate complexes act as catalysts in 3-*exo*-(dimethylamino)isonorborneol (DAIB) promoted dialkylzinc additions to aldehydes.⁹ These monomers equilibrate with their dimers. X-ray structures show that the Zn–Me groups in homochiral DAIB dimers are *syn* (as shown in Scheme 2) while heterochiral DAIB dimers have *anti* alignments of Zn–Me moieties. The *syn* alignments contribute to a lower stability of the homochiral dimers and provide the basis for a higher catalyst reactivity compared to heterochiral dimers. Chirality amplification phenomena are also influenced by these differences.^{6a–c,9}

We here present X-ray crystal structures of two chiral organometallic complexes, which have been obtained by treating the chiral chelating alkoxy ligand (1*R*,2*R*,4*S*)-2-*endo*-hydroxy-2-*exo*-(*o*-methoxybenzene)-1,3,3-trimethylbicyclo[2.2.1]heptane (**1**)¹⁰ with two frequently used organometallic reagents, *n*-butyllithium and dimethylzinc.

Treatment of **1** with *n*-butyllithium in hexanes at 0 °C and recrystallization from toluene/hexanes yields colorless crystals of **2**.¹¹ X-ray crystal analysis of **2** reveals a distorted cubic Li₄C₁O₃ core, consistent with three {lithium (1*R*,2*R*,4*S*)-2-*endo*-oxido-2-*exo*-(*o*-methoxybenzene)-1,3,3-trimethylbicyclo[2.2.1]heptane} (**1-Li**)



moieties and one *n*-butyllithium subunit (Figure 1).¹² The oxygen atoms of the three methoxy groups coordinate the lithiums of the Li₄C₁O₃ core in a special way: the three Li ions, which are coordinated to the C_α atom, are chelated by oxido and methoxy groups, the fourth lithium (Li_{trans}) by only three alkoxide ions (Figure 1). The structure does not allow tetracoordination of all four Li cations. Li_{trans} appears less attractive for coordination due to its sterically crowded environment.¹³ It is coordinated to three bulky alkoxido units, while each Li_{cis} ion contacts two alkoxide moieties and the smaller *n*-butyl group. The *n*-butyl group in **2** adopts a nearly perfect antiperiplanar conformation, the C_α–C_β–C_γ–C_δ dihedral angle is 177°. The methyl groups of the methoxy units are positioned out of the planes of the attached benzene rings by 19.4°, 19.7°, and 32.6°. ¹⁴

Reaction of dimethylzinc with a hexane solution of **1** at 0 °C and recrystallization from hexanes yield colorless crystals of **3**.¹⁵ The X-ray structure of **3** reveals a C₂

(11) Synthesis of **2**: 1.2 mmol *n*-butyllithium (1.6 M in hexanes, 0.75 mL) were added at 0 °C to a solution of 0.9 mmol (234 mg) **1** in 1 mL of hexanes. A white precipitate formed and was dissolved in hot toluene/hexanes. Colorless crystals of **2** were obtained by cooling this solution to room temperature (39% yield). X-ray crystallography revealed that **2** was also formed (in lower yield) if a 1:1 ratio of *n*-butyllithium to **1** was employed. X-ray crystal data of **2**: C₅₅H₇₈Li₄O₆; *M* = 862.93; orthorhombic; space group *P*2₁2₁2₁; *a* = 19.150(16) Å; *b* = 18.387(13) Å; *c* = 13.603(10) Å; *V* = 4789.6(62) Å³; *Z* = 4; *T* = 148(2) K; *μ* = 0.074; reflections total, 7304; reflections observed (>2σ), 3422; parameters refined, 606; final *R* values, *R*_{obs} = 0.0606; w*R*_{all} = 0.182; GOF_{all} = 0.999. The absolute configuration originates from (–)-fenchone, (1*R*,4*S*)-1,3,3-trimethylbicyclo[2.2.1]heptane-2-one.

(12) To our knowledge, this is the first example of a cubic mixed anion Li₄C₁O₃ cluster; Cambridge Structural Database (CSD), 1998.

(13) Higher than three-coordination of Li_{trans} in **2** is prevented by sterically demanding alkyl groups, which imbed Li_{trans}. Shortest Li_{trans}–C, H distances are 3.14 and 2.52 Å, respectively, and point to the tendency to form “agostic” Li–HC interactions. For short Li–C(H) distances, see: (a) Goldfuss, B.; Schleyer, P. v. R.; Hampel, F. *J. Am. Chem. Soc.* **1996**, *118*, 12183. (b) Kottke, T.; Stalke, D. *Angew. Chem.* **1993**, *105*, 619; *Angew. Chem., Int. Ed. Engl.* **1993**, *32*, 580.

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(15) Synthesis of **3**: 0.4 mmol (0.4 mL) of dimethylzinc (0.9 M in hexanes) were added at 0 °C to a solution of 0.4 mmol (104 mg) of **1** in 1 mL of hexanes. The mixture was stirred (methane evolved) for 1 h. A white precipitate formed and was dissolved in hot hexanes. Colorless crystals of **3** were obtained by cooling this solution to room temperature (58% yield). X-ray crystal data of **3**: C₃₆H₅₂O₄Zn₂; *M* = 679.52; orthorhombic; space group *P*2₁2₁2₁; *a* = 13.405(6) Å; *b* = 13.508(6) Å; *c* = 9.233(4) Å; *V* = 1671.9(13) Å³; *Z* = 2; *T* = 148(2) K; *μ* = 1.471; reflections total, 2758; reflections observed (>2σ), 2308; parameters refined, 244; final *R* values, *R*_{obs} = 0.0319; w*R*_{all} = 0.0853; GOF_{all} = 1.060. The absolute configuration originates from (–)-fenchone, (1*R*,4*S*)-1,3,3-trimethylbicyclo[2.2.1]heptane-2-one.

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(10) Ligand **1** has been synthesized from (–)-fenchone and *o*-lithioanisole. For the enantiomer of **1**, (1*S*,2*S*,4*R*)-2-*endo*-hydroxy-2-*exo*-(*o*-methoxybenzene)-1,3,3-trimethylbicyclo[2.2.1]heptane, see: (a) Starling, S. M.; Vonwiller, S. C.; Reek, J. N. H. *J. Org. Chem.* **1998**, *63*, 2262. (b) Fry, J. L.; West, J. W. *J. Org. Chem.* **1981**, *46*, 2177.

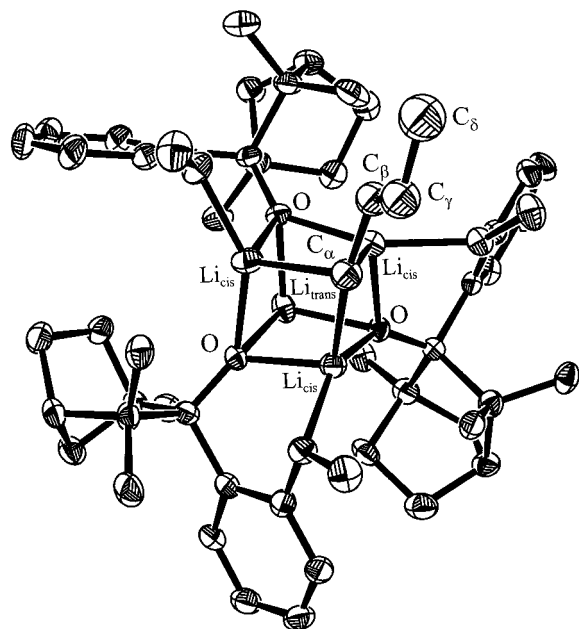


Figure 1. X-ray crystal structure of **2**. Hydrogen atoms are omitted for clarity.

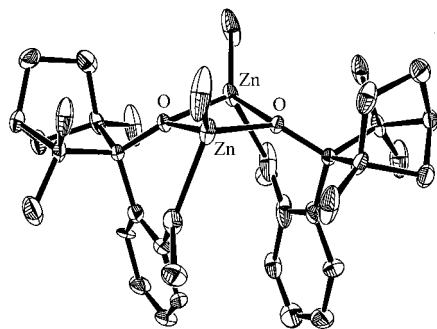


Figure 2. X-ray crystal structure of **3**. Disordered aryl and methoxy methyl groups are not shown. Hydrogen atoms are omitted for clarity.

symmetric, dimeric aggregate of the methyl zinc chelate complex **3** (Figure 2). Out of the three possible isomers with different configurations at the tetrahedral Zn centers, namely *R,R*, *R,S*, and *S,S*, complex **3** adopts *S,S* configurations at the Zn centers. As in Noyori's homochiral DAIB methylzinc dimer (Scheme 2),^{9d} the two methyl groups at the zinc atoms in **3** are aligned

syn with respect to the central Zn_2O_2 ring (Figure 2). The bond distances in the central Zn_2O_2 ring (2.012 Å for the dimer forming Zn–O bond and 1.966 Å for the internal Zn–O bond) are both shorter in **3** than in Noyori's homochiral dimer (2.05 and 1.98 Å, respectively).^{9d} The central Zn_2O_2 ring in **3** is not planar, but puckered significantly (O–Zn–O–Zn dihedral angle: 16°). The planarity (bond angle sum) at the 3-fold coordinated O atom in the Zn_2O_2 ring is higher in **3** (353°) than in Noyori's homochiral dimer (347° and 341°).^{9d} Disordered aryl CH and methoxy CH_3 groups are apparent in **3**. The methoxy methyl groups bend by 31.9° and (due to disorder) by 26.1° out of the plane of the aryl moieties.¹⁴ The folded Zn_2O_2 conformation in **3** supports close contacts between C_{meta} and C_{para} atoms of the aryl groups (2.981 and 3.152 Å). However, complete π – π interactions involving all aryl C ring atoms are not apparent in **3**, due to a skew alignment of the aryl groups (Figure 2).

Our studies demonstrate that the chiral chelating ligand **1** readily forms with *n*-butyllithium the chiral mixed aggregate **2**. This implies the possibility of complexation with other organolithiums (e.g., *s*- or *t*-BuLi) and suggests their use in enantioselective reactions with prochiral electrophiles. The close structural similarities between dimeric **3** and Noyori's highly efficient catalyst (Scheme 2) point to promising applications of **3** as catalyst in enantioselective additions of alkylzincs to aldehydes. The suitabilities of **2** and **3** for enantioselective reactions are currently under investigation.

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Supporting Information Available: Tables of X-ray structural data for compounds **2** and **3**. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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